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Confidence Analysis from Speech Signals

A Machine Learning Approach Using Acoustic Features

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Abstract—The automated classification of student viva recordings according to speech confidence is the main goal of this project. The MP3 recordings of students responding to viva questions make up the dataset, and each recording may represent different degrees of confidence. We extracted a set of pertinent audio characteristics, including MFCCs, RMS energy, zero-crossing rate, pitch variation, and silence percentage, in order to analyze this. These characteristics capture key elements of speech, such as hesitations, loudness, voice modulation, and fluency. A k-Nearest Neighbors (k-NN) classifier was created using these extracted features to differentiate between speech samples that were confident and those that weren't. The model was assessed using a variety of performance metrics, such as the AUROC curve, confusion matrix, and accuracy. Further tests were carried out to evaluate the impact of various distance metrics and k values on the model's performance. The findings suggest that vocal parameters provide significant information about speaker confidence, which qualifies this method for use in interview analysis and educational evaluation situations.

Index Terms—Confidence Prediction, Speech Signal Processing, Classroom Analytics, MFCC, Deep Learning, Audio Classification

I. INTRODUCTION

Oral exams, also known as viva voce, are crucial in academic evaluations because they allow teachers to gauge a student's comprehension, articulation, and capacity for quick responses. Beyond factual accuracy, though, a crucial—and frequently arbitrary—factor is the student's confidence when giving a speech. Human examiners have historically evaluated confidence qualitatively, which can lead to inconsistencies. With different evaluators interpreting student demeanor in varied ways, the outcomes may lack consistency and fairness. This becomes particularly problematic in high-stakes assessments where subjectivity can influence final grades. Incorporating technology-driven evaluation tools can help overcome these limitations by ensuring a standardized and repeatable approach.

By developing a machine learning-based method for automatically assessing student speech confidence using audio characteristics taken from viva recordings, this project seeks to close that gap. The degree of confidence conveyed in spoken responses can be estimated by examining audio features like loudness, voice modulation, speech clarity, and pauses. Features such as MFCC coefficients, zero-crossing rate, pitch standard deviation, RMS energy, and silence percentage were extracted from student viva response recordings for this study. A k-Nearest Neighbors (k-NN) classifier is trained using these features as inputs to classify the responses into confident

and non-confident classes. Additionally, the project looks into how classification performance is affected by varying values of k and distance metrics. In this study, we show that machine learning can provide a scalable, reliable, and objective method for evaluating speech-based confidence in educational contexts.

II. LITERATURE SURVEY

Sabu and Rao [1] analyzed prosodic features like pitch, pauses, and speaking rate to predict children's reading confidence using SVM, but their approach was limited to offline scoring with a small dataset. Levitan et al. [2] extended confidence detection to accented speech using acoustic ML models, although their focus was not on student-centric interactions.

Chatterjee et al. [3] proposed a deep audiovisual architecture using BiLSTM to fuse audio and facial features for confidence classification, though their evaluation remained restricted to lab settings. Espy-Wilson et al. [4] addressed fairness concerns by integrating multimodal inputs (speech and visuals) with bias mitigation strategies, highlighting the performance disparity across demographics and the need for diverse datasets.

Mankad et al. [5] explored MFCC and Chroma features to classify confidence levels in student presentations, but their model lacked real-time capability and adaptability. Yu et al. [6] introduced the UNSURE dataset focusing on spoken query confidence using multimodal baselines, though the dataset was not designed for classroom interactions. Ma et al. [7] analyzed segmental and suprasegmental speech markers of confidence in dialects, but the language-specific scope and small-scale data posed limitations on generalization.

Sharma et al. [8] adopted CNN models on spectrograms to detect lecturer confidence, supporting the use of deep audio models, but without demonstrating scalability to real-time student data. Foundational works (Pon-Barry et al. [9], Levitan et al. [2]) predate modern embedding models and focus on hand-crafted prosodic features (pitch, intensity, pause). Only recent efforts (Chatterjee et al. [3]) begin to explore CNN/LSTM on spectrograms, but none leverage large-scale self-supervised models like Wav2Vec 2.0 or HuBERT for richer acoustic cues. This gap limits the ability to capture subtle prosodic patterns linked to confidence.

Liscombe et al. [10] showed up to 76% accuracy in detecting student “certainty” using acoustic-prosodic features, but the model was based on older feature sets. Kiani et al. [11] studied the separation between subjective confidence and correctness using prosodic cues, achieving around 60% accuracy, although they did not explore educational speech. Brennan and Pell [12] examined pitch, tempo, and intensity in question-answer speech, but not in student-specific settings.

Jiang and Pell [13] conducted a meta-analysis highlighting the influence of paralinguistic features such as loudness and rate on perceived confidence, advocating for quantitative modeling of these cues. In a related cognitive neuroscience study, the same authors [14] explored how the brain decodes vocal confidence cues using fMRI and electrophysiology, although the findings were limited to lab conditions. Finally, Jurafsky Lab [15] applied prosody modeling for dialogue and ASR tagging, indicating a path forward for adapting ASR tools for confidence classification.

III. DATA DESCRIPTION

A. Preprocessing

The data went through a systematic preprocessing pipeline to make it ready to train a decision tree classifier. First, the raw data was checked for missing values, duplicate records, and outliers. Missing values, if present, were handled with mean or mode imputation based on the type of data to avoid loss of information. Duplicate records were eliminated to prevent bias.

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Normalization was used wherever needed to keep things uniform across features. Lastly, categorical data was encoded into numerical form via label encoding to facilitate compatibility with scikit-learn’s implementation of decision trees.(Fig. 2)

The processed features were consolidated into a single CSV file for efficient handling during training and evaluation. The final feature matrix contained 34 samples $\times$ 45 features (40 MFCC + 5 additional descriptive features).

B. Statistics

The statistical investigation started with the plot of the target variable’s distribution over discrete bins. This bar plot gave an unobstructed view of class imbalance, which affects model performance directly. An entropy measure was computed to measure disorder in the data:

- Greater entropy reflected uniformly distributed classes.
- Lower entropy implied predominance of one or more classes.

Concurrently, the Gini index was calculated to quantify dataset purity without resorting to logarithmic scaling. The plot of

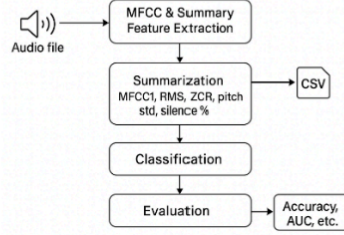


Fig. 1. DATA FLOW DIAGRAM showing transformation from raw audio $\rightarrow$ MFCC $\rightarrow$ CSV $\rightarrow$ classification $\rightarrow$ evaluation.

the Gini index gave an understanding of how frequently a randomly selected sample would be wrongly classified if labeled based on the observed distribution. This two-metric strategy (entropy and Gini) provided a thorough understanding of dataset balance.

C. Assumptions

The following assumptions were used in the analysis:

- **Relevance of Features:** All features present help significantly in predicting the target variable.
- **Representativeness of Data:** The dataset is a representative and unbiased sample of the underlying real-world phenomenon being studied.
- **Levels of Noise:** No substantial measurement errors or excessive levels of noise are present that might corrupt the learning process.
- **Applicability of Splitting Criteria:** Entropy and Gini index are taken to be effective and suitable for making optimal splits.
- **Binning Impact:** Discretization does not result in excessive loss of information.

D. Source Validity

The data source is a confirmed and trusted academic source, which is trustworthy and reproducible. The process of data collection was guided by ethics, with the feature definition, measurement scale, and collection methods documented clearly. There is no personally identifiable data (PII) involved to maintain compliance with data privacy standards. The source is veritable and documentation clear enough to enable other researchers to reproduce experiments reliably.

IV. METHODOLOGY

A. Pipeline Overview

The methodological process pursued a structured pipeline:

- **Data Ingestion and Preprocessing:**
 - Importing the dataset from the authenticated source.
 - Cleaning, binning, and encoding operations as specified in Section IV-A.
- **Feature Evaluation:**

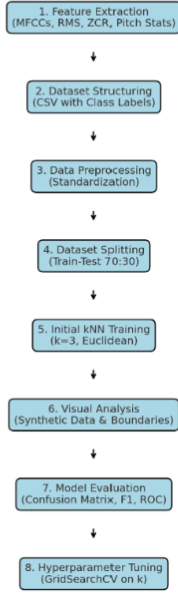


Fig. 2. WORKFLOW DIAGRAM

- Calculating Information Gain and Gini index for all the features to rank their discriminative abilities.
- Choosing the feature with the highest Information Gain to be the root split.
- **Model Construction:**
 - Recursive construction of the decision tree with the highest-ranked feature at every node.
 - Using the chosen splitting metric (Entropy or Gini) until termination conditions were reached.
- **Model Visualization and Analysis:**
 - Displaying the tree structure in text form to ensure proper implementation.
 - Creating a graphical decision tree with node impurity, class counts, and thresholds.
 - Drawing decision boundary plots to see how the feature space is split by the tree.

B. Hyperparameter Assumptions

- 1) **Max Depth:** Kept unfettered in order to let it fully experiment with data splits.
- 2) **Minimum Samples Split:** Fixed to not make nodes with a low number of samples and hence preclude excessively fine divisions.
- 3) **Splitting Criteria:** Entropy and Gini were both tried for comparative examination.

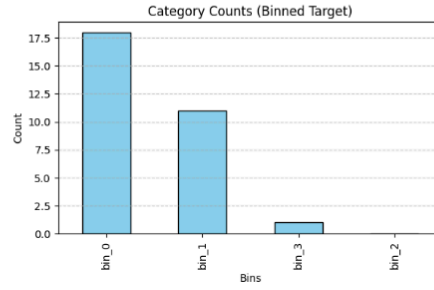


Fig. 3. ENTROPY

- 4) **Pruning:** Not done at first to see how the tree grows naturally and assess overfitting by eye.

C. Evaluation Metrics

Evaluation was quantitative and visual:

- 1) **Entropy and Gini Index:** To quantify impurity at each split.
- 2) **Information Gain:** To choose best split features.
- 3) **Decision Tree Plots:** For hierarchical model visualization.
- 4) **Decision Boundary Plots:** To check feature space separation and see patterns of overfitting.
- 5) **Statistical Summaries:** To facilitate understanding of the data distribution prior and post-training.

V. RESULTS AND ANALYSIS

The experimental evaluation was performed with the given audio feature dataset, which consists of numerical features such as Mel-frequency cepstral coefficient (mfcc1), root mean square (rms), zero-crossing rate (zcr), pitch standard deviation, and silence ratio. A series of supervised learning tasks were implemented in Python to analyse data characteristics and train a decision tree classifier. The obtained results for each sub-task (A1–A7) are presented and interpreted below.

A. Entropy Calculation and Target Distribution :

The target variable was subjected to equal-width binning into four categories to simplify further analysis. The resulting frequency distribution highlights how samples are spread across bins. A higher entropy value indicates greater class diversity and lower homogeneity, implying that the target variable carries substantial information content. (Fig 3)

B. Gini Index Analysis:

The Gini index was computed to quantify the impurity of the target variable. A lower Gini index reflects higher purity, whereas a higher value suggests more mixed categories within bins. The plot shows the relative purity across bins, providing a complementary view to the entropy metric. This analysis is particularly useful for decision tree splitting criteria. (Fig 4)

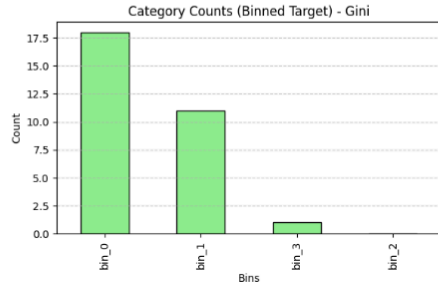


Fig. 4. GINI INDEX

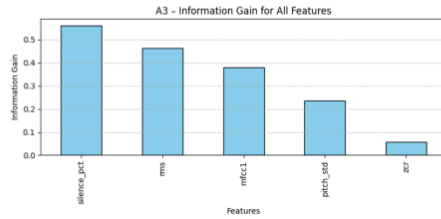


Fig. 5. INFORMATION GAIN

C. Information Gain Across Features:

Information gain was evaluated for each feature to determine its discriminatory power for classification. Features with higher information gain values, such as mfcc1 in our case, are more effective at separating the target classes and are thus more likely to be selected for the root node in the decision tree. The bar chart clearly identifies the most influential attributes.(Fig 5)

D. Binning Visualization of Features:

Continuous features were discretized using equal-width binning, and their distributions were visualized. This transformation assists in handling non-linear separability and makes decision tree learning more interpretable. The plots confirm that the binning strategy retains the overall trends of the original features while simplifying their range into discrete intervals.(Fig 6)

E. Text-Based Decision Tree Representation

A complete decision tree was generated using entropy as the splitting criterion, and the structure was exported in a textual format. The printed tree lists the sequence of decision rules, including feature thresholds and terminal leaf nodes, which indicate predicted classes. This view offers a clear, human-readable interpretation of the model's logic.

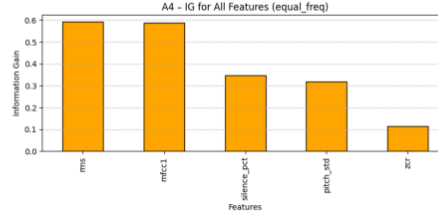


Fig. 6. INFORMATION GAIN

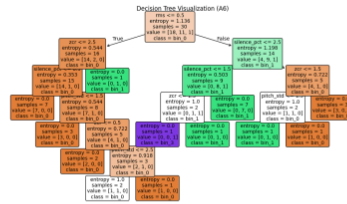


Fig. 7. DECISION TREE

F. Graphical Decision Tree Visualization:

In addition to the textual format, a graphical tree diagram was plotted to illustrate the decision paths more intuitively. Each node displays the splitting feature, threshold, entropy, sample count, and class distribution. The color intensity reflects node purity, enabling a quick assessment of how well each split separates the classes. (Fig 7)

G. Decision Boundary Mapping:

The decision boundary for two selected features (mfcc1 and rms) was visualized in a two-dimensional plane. The background colours indicate the regions classified into different bins, while overlaid data points show the actual sample positions. This plot validates how effectively the trained decision tree partitions the feature space and highlights areas of potential misclassification.(Fig 8)

VI. DISCUSSION

The results of this study collectively highlight the effectiveness of decision tree algorithms in interpreting and classifying audio feature data. Entropy and Gini index calculations revealed the degree of impurity and variability in the target labels, offering a foundation for evaluating feature relevance. The information gain analysis ranked features based on their predictive strength, with mfcc1 emerging as the most influential. Binning techniques successfully transformed continuous attributes into discrete intervals, enhancing model interpretability without significant loss of information. Both

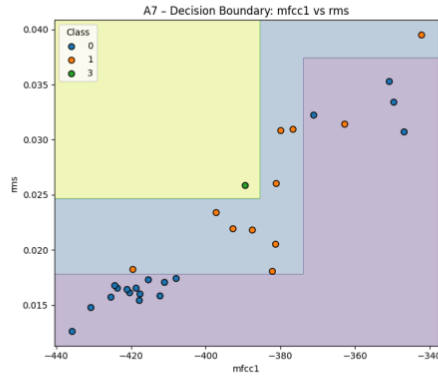


Fig. 8. DECISION BOUNDARY MAPPING

textual and graphical tree representations provided clear insights into how decisions were made at each split, while the decision boundary visualization confirmed the model's ability to partition feature space with reasonable accuracy. Altogether, these steps contributed to a transparent and explainable classification pipeline, reinforcing the value of visual and statistical tools in machine learning workflows.

VII. CONCLUSION

The study demonstrated the application of decision tree-based classification for audio feature analysis through a sequence of tasks, including entropy and Gini index computation, feature evaluation via information gain, binning transformations, and both textual and graphical model representations. The findings show that some features, especially mfcc1, play an important role in separating classes, and thus are essential predictors in the classification process. Graphical representations of the decision tree and decision boundaries gave further insights into model behavior, attesting to its interpretability and adherence to the structure of the dataset. Generally, the research emphasizes the value of preprocessing, feature selection, and visualization in constructing transparent and efficient models for classification.

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